

src/components/ChangedFilesTree.tsx

Kind: added · Baseline: a9c34bfb03fe · Target: NovaDiff · Generated: 2026-05-20T04:51:01.354Z

Overview

`src/components/ChangedFilesTree.tsx` is a new component that renders a virtualized tree of changed files. It uses `react-window` for efficient scrolling and manages directory expansion, selection, and dynamic sizing.

Key changes

- **Imports** (lines 1-11):
 - `FixedSizeList` and `ListChildComponentProps` from `react-window`.
 - `FileChange` type and tree-building helpers (`ancestorDirPaths`, `buildChangedFilesTreeRoot`, `flattenVisibleRows`, `FlatRow`) from `../app/changedFilesTree`.
- **ChangedFilesTreeProps interface** (lines 20-24) defines `rows`, `selectedPath`, and `onSelectFile`.
- **TreeRow component** (lines 34-88) renders each row, handling directory toggles and file selection with ARIA attributes.
- **ChangedFilesTree component** (lines 90-217) sets up refs, state for list dimensions, and memoized data for the list. It uses `useDeferredValue` to avoid blocking UI on rapid row changes.
- **Layout effects:**
 - Reset expanded dirs when `rows` change (lines 111-117).
 - Expand ancestor dirs when `selectedPath` changes (lines 131-150).
 - Observe host size with `ResizeObserver` to update `listWidth` / `listHeight` (lines 152-167).
 - Scroll to the selected file when it appears (lines 169-179).
- **Conditional rendering:** shows messages when no rows or no visible files (lines 192-196, 200-202).

Impact

- **Performance:** Virtualized list reduces DOM nodes for large diffs.
- **UX:** Auto-expansion of ancestor dirs and scrolling to the selected file improve navigation.
- **Dependencies:** Requires `react-window`; ensure it is installed and bundled.
- **Styling:** New CSS classes (`file-tree-virt-row`, `file-tree-dir`, `file-pill`) must exist or be added.
- **Testing:** Existing tests for file selection and directory toggling need updates to cover the new component.

Risks & follow-ups

- Verify `ResizeObserver` cleanup works across browsers (Safari/Edge).
- Confirm `useDeferredValue` does not cause stale UI when `rows` update rapidly.
- Ensure `TreeRow` handles missing `row` gracefully (currently returns `null`).
- Run lint, unit, and integration tests to catch any missing imports or type errors introduced by the new file.